UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

x **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023

OR

o **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from _________ to _________

Commission File Number: 001-34756

Tesla, Inc.

(Exact name of registrant as specified in its charter)

Delaware	**91-2197729**
(State or other jurisdiction of incorporation or organization)	**(I.R.S. Employer Identification No.)**
1 Tesla Road Austin, Texas	**78725**
(Address of principal executive offices)	**(Zip Code)**

(512) 516-8177

(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock	TSLA	The Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act:

None

Indicate by check mark whether the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes x No o

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or 15(d) of the Act. Yes o No x

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 ("Exchange Act") during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes x No o

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes x No o

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act:

Large accelerated filer	x	Accelerated filer	o
Non-accelerated filer	o	Smaller reporting company	o
Emerging growth company	o		

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. o

Indicate by check mark whether the Registrant has filed a report on and attestation to its management’s assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. x

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. o

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant’s executive officers during the relevant recovery period pursuant to §240.10D-1(b). o

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes o No x

The aggregate market value of voting stock held by non-affiliates of the registrant, as of June 30, 2023, the last day of the registrant’s most recently completed second fiscal quarter, was $722.52 billion (based on the closing price for shares of the registrant’s Common Stock as reported by the NASDAQ Global Select Market on June 30, 2023). Shares of Common Stock held by each executive officer and director have been excluded in that such persons may be deemed to be affiliates. This determination of affiliate status is not necessarily a conclusive determination for other purposes.

As of January 22, 2024, there were 3,184,790,415 shares of the registrant’s common stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant’s Proxy Statement for the 2024 Annual Meeting of Stockholders are incorporated herein by reference in Part III of this Annual Report on Form 10-K to the extent stated herein. Such proxy statement will be filed with the Securities and Exchange Commission within 120 days of the registrant’s fiscal year ended December 31, 2023.

TESLA, INC.

ANNUAL REPORT ON FORM 10-K FOR THE YEAR ENDED DECEMBER 31, 2023

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Forward-Looking Statements

The discussions in this Annual Report on Form 10-K contain forward-looking statements reflecting our current expectations that involve risks and uncertainties. These forward-looking statements include, but are not limited to, statements concerning supply chain constraints, our strategy, competition, future operations and production capacity, future financial position, future revenues, projected costs, profitability, expected cost reductions, capital adequacy, expectations regarding demand and acceptance for our technologies, growth opportunities and trends in the markets in which we operate, prospects and plans and objectives of management. The words “anticipates,” “believes,” “could,” “estimates,” “expects,” “intends,” “may,” “plans,” “projects,” “will,” “would” and similar expressions are intended to identify forward-looking statements, although not all forward-looking statements contain these identifying words. We may not actually achieve the plans, intentions or expectations disclosed in our forward-looking statements and you should not place undue reliance on our forward-looking statements. Actual results or events could differ materially from the plans, intentions and expectations disclosed in the forward-looking statements that we make. These forward-looking statements involve risks and uncertainties that could cause our actual results to differ materially from those in the forward-looking statements, including, without limitation, the risks set forth in Part I, Item 1A, “Risk Factors” of the Annual Report on Form 10-K for the fiscal year ended December 31, 2023 and that are otherwise described or updated from time to time in our other filings with the Securities and Exchange Commission (the “SEC”). The discussion of such risks is not an indication that any such risks have occurred at the time of this filing. We do not assume any obligation to update any forward-looking statements.

PART I

ITEM 1. BUSINESS

Overview

We design, develop, manufacture, sell and lease high-performance fully electric vehicles and energy generation and storage systems, and offer services related to our products. We generally sell our products directly to customers, and continue to grow our customer-facing infrastructure through a global network of vehicle showrooms and service centers, Mobile Service, body shops, Supercharger stations and Destination Chargers to accelerate the widespread adoption of our products. We emphasize performance, attractive styling and the safety of our users and workforce in the design and manufacture of our products and are continuing to develop full self-driving technology for improved safety. We also strive to lower the cost of ownership for our customers through continuous efforts to reduce manufacturing costs and by offering financial and other services tailored to our products.

Our mission is to accelerate the world’s transition to sustainable energy. We believe that this mission, along with our engineering expertise, vertically integrated business model and focus on user experience differentiate us from other companies.

Segment Information

We operate as two reportable segments: (i) automotive and (ii) energy generation and storage.

The automotive segment includes the design, development, manufacturing, sales and leasing of high-performance fully electric vehicles as well as sales of automotive regulatory credits. Additionally, the automotive segment also includes services and other, which includes sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise. The energy generation and storage segment includes the design, manufacture, installation, sales and leasing of solar energy generation and energy storage products and related services and sales of solar energy systems incentives.

Our Products and Services

Automotive

We currently manufacture five different consumer vehicles – the Model 3, Y, S, X and Cybertruck. Model 3 is a four-door mid-size sedan that we designed for manufacturability with a base price for mass-market appeal. Model Y is a compact sport utility vehicle (“SUV”) built on the Model 3 platform with seating for up to seven adults. Model S is a four-door full-size sedan and Model X is a mid-size SUV with seating for up to seven adults. Model S and Model X feature the highest performance characteristics and longest ranges that we offer in a sedan and SUV, respectively. In November 2023, we entered the consumer pickup truck market with first deliveries of the Cybertruck, a full-size electric pickup truck with a stainless steel exterior that has the utility and strength of a truck while featuring the speed of a sports car.

In 2022, we also began early production and deliveries of a commercial electric vehicle, the Tesla Semi. We have planned electric vehicles to address additional vehicle markets, and to continue leveraging developments in our proprietary Full Self-Driving (“FSD”) Capability features, battery cell and other technologies.

Energy Generation and Storage

Energy Storage Products

Powerwall and Megapack are our lithium-ion battery energy storage products. Powerwall, which we sell directly to customers, as well as through channel partners, is designed to store energy at a home or small commercial facility. Megapack is an energy storage solution for commercial, industrial, utility and energy generation customers, multiple of which may be grouped together to form larger installations of gigawatt hours (“GWh”) or greater capacity.

We also continue to develop software capabilities for remotely controlling and dispatching our energy storage systems across a wide range of markets and applications, including through our real-time energy control and optimization platforms.

Solar Energy Offerings

We sell retrofit solar energy systems to customers and channel partners and also make them available through power purchase agreement ("PPA") arrangements. We purchase most of the components for our retrofit solar energy systems from multiple sources to ensure competitive pricing and adequate supply. We also design and manufacture certain components for our solar energy products.

We sell our Solar Roof, which combines premium glass roof tiles with energy generation, directly to customers, as well as through channel customers. We continue to improve our installation capability and efficiency, including through collaboration with real estate developers and builders on new homes.

Technology

Automotive

Battery and Powertrain

Our core vehicle technology competencies include powertrain engineering and manufacturing and our ability to design vehicles that utilize the unique advantages of an electric powertrain. We have designed our proprietary powertrain systems to be adaptable, efficient, reliable and cost-effective while withstanding the rigors of an automotive environment. We offer dual motor powertrain vehicles, which use two electric motors to maximize traction and performance in an all-wheel drive configuration, as well as vehicle powertrain technology featuring three electric motors for further increased performance in certain versions of Model S and Model X, Cybertruck and the Tesla Semi.

We maintain extensive testing and R&D capabilities for battery cells, packs and systems, and have built an expansive body of knowledge on lithium-ion cell chemistry types and performance characteristics. In order to enable a greater supply of cells for our products with higher energy density at lower costs, we have developed a new proprietary lithium-ion battery cell and improved manufacturing processes.

Vehicle Control and Infotainment Software

The performance and safety systems of our vehicles and their battery packs utilize sophisticated control software. Control systems in our vehicles optimize performance, customize vehicle behavior, manage charging and control all infotainment functions. We develop almost all of this software, including most of the user interfaces, internally and update our vehicles' software regularly through over-the-air updates.

Self-Driving Development and Artificial Intelligence

We have expertise in developing technologies, systems and software to enable self-driving vehicles using primarily vision-based technologies. Our FSD Computer runs our neural networks in our vehicles, and we are also developing additional computer hardware to better enable the massive amounts of field data captured by our vehicles to continually train and improve these neural networks for real-world performance.

Currently, we offer in our vehicles certain advanced driver assist systems under our Autopilot and FSD Capability options. Although at present the driver is ultimately responsible for controlling the vehicle, our systems provide safety and convenience functionality that relieves drivers of the most tedious and potentially dangerous aspects of road travel much like the system that airplane pilots use, when conditions permit. As with other vehicle systems, we improve these functions in our vehicles over time through over-the-air updates.

We intend to establish in the future an autonomous Tesla ride-hailing network, which we expect would also allow us to access a new customer base even as modes of transportation evolve.

We are also applying our artificial intelligence learnings from self-driving technology to the field of robotics, such as through Optimus, a robotic humanoid in development, which is controlled by the same AI system.

Energy Generation and Storage

Energy Storage Products

We leverage many of the component-level technologies from our vehicles in our energy storage products. By taking a modular approach to the design of battery systems, we can optimize manufacturing capacity of our energy storage products. Additionally, our expertise in power electronics enables our battery systems to interconnect with electricity grids while providing fast-acting systems for power injection and absorption. We have also developed software to remotely control and dispatch our energy storage systems.

Solar Energy Systems

We have engineered Solar Roof over numerous iterations to combine aesthetic appeal and durability with power generation. The efficiency of our solar energy products is aided by our own solar inverter, which incorporates our power electronics technologies. We designed both products to integrate with Powerwall.

Design and Engineering

Automotive

We have established significant in-house capabilities in the design and test engineering of electric vehicles and their components and systems. Our team has significant experience in computer-aided design as well as durability, strength and crash test simulations, which reduces the product development time of new models. We have also achieved complex engineering feats in stamping, casting and thermal systems, and developed a method to integrate batteries directly with vehicle body structures without separate battery packs to optimize manufacturability, weight, range and cost characteristics.

We are also expanding our manufacturing operations globally while taking action to localize our vehicle designs and production for particular markets, including country-specific market demands and factory optimizations for local workforces. As we increase our capabilities, particularly in the areas of automation, die-making and line-building, we are also making strides in the simulations modeling these capabilities prior to construction.

Energy Generation and Storage

Our expertise in electrical, mechanical, civil and software engineering allows us to design, engineer, manufacture and install energy generating and storage products and components, including at the residential through utility scale. For example, the modular design of our Megapack utility-scale battery line is intended to significantly reduce the amount of assembly required in the field. We also customize solutions including our energy storage products, solar energy systems and/or Solar Roof for customers to meet their specific needs.

Sales and Marketing

Historically, we have been able to achieve sales without traditional advertising and at relatively low marketing costs. We continue to monitor our public narrative and brand, and tailor our marketing efforts accordingly, including through investments in customer education and advertising as necessary.

Automotive

Direct Sales

Our vehicle sales channels currently include our website and an international network of company-owned stores. In some jurisdictions, we also have galleries to educate and inform customers about our products, but such locations do not transact in the sale of vehicles. We believe this infrastructure enables us to better control costs of inventory, manage warranty service and pricing, educate consumers about electric vehicles, make our vehicles more affordable, maintain and strengthen the Tesla brand and obtain rapid customer feedback.

We reevaluate our sales strategy both globally and at a location-by-location level from time to time to optimize our sales channels. However, sales of vehicles in the automobile industry tend to be cyclical in many markets, which may expose us to volatility from time to time.

Used Vehicle Sales

Our used vehicle business supports new vehicle sales by integrating the trade-in of a customer's existing Tesla or non-Tesla vehicle with the sale of a new or used Tesla vehicle. The Tesla and non-Tesla vehicles we acquire as trade-ins are subsequently remarketed, either directly by us or through third parties. We also remarket used Tesla vehicles acquired from other sources including lease returns.

Public Charging

We have a growing global network of Tesla Superchargers, which are our industrial-grade, high-speed vehicle chargers. Where possible, we co-locate Superchargers with our solar and energy storage systems to reduce costs and promote renewable power. Supercharger stations are typically placed along well-traveled routes and in and around dense city centers to allow vehicle owners the ability to enjoy quick, reliable charging along an extensive network with convenient stops. Use of the Supercharger network either requires payment of a fee or is free under certain sales programs. In November 2021, we began to offer Supercharger access to non-Tesla vehicles in certain locations in support of our mission to accelerate the world's transition to sustainable energy, and in November 2022, we opened up our previously proprietary charging connector as the North American Charging Standard (NACS). This enables all electric vehicles and charging stations to interoperate — which makes charging easier and more efficient for everyone and advances our mission to accelerate the world's transition to sustainable energy. Following this, a number of major automotive companies announced their adoption of NACS, with their access to the Supercharger network beginning in phases in 2024 and their production of NACS vehicles beginning no later than 2025. We also engaged SAE International to govern NACS as an industry standard, now named J3400. We continue to monitor and increase our network of Tesla Superchargers in anticipation of future demand.

We also work with a wide variety of hospitality, retail and public destinations, as well as businesses with commuting employees, to offer additional charging options for our customers, as well as single-family homeowners and multi-family residential entities, to deploy home charging solutions.

In-App Upgrades

As our vehicles are capable of being updated remotely over-the-air, our customers may purchase additional paid options and features through the Tesla app or through the in-vehicle user interface. We expect that this functionality will also allow us to offer certain options and features on a subscription basis in the future.

Energy Generation and Storage

We market and sell our solar and energy storage products to residential, commercial and industrial customers and utilities through a variety of channels, including through our website, stores and galleries, as well as through our network of channel partners, and in the case of some commercial customers, through PPA transactions. We emphasize simplicity, standardization and accessibility to make it easy and cost-effective for customers to adopt clean energy, while reducing our customer acquisition costs.

Service and Warranty

Automotive

Service

We provide service for our electric vehicles at our company-owned service locations and through Tesla Mobile Service technicians who perform work remotely at customers' homes or other locations. Servicing the vehicles ourselves allows us to identify problems and implement solutions and improvements faster than traditional automobile manufacturers and their dealer networks. The connectivity of our vehicles also allows us to diagnose and remedy many problems remotely and proactively.

Vehicle Limited Warranties and Extended Service Plans

We provide a manufacturer's limited warranty on all new and used Tesla vehicles we sell directly to consumers, which may include limited warranties on certain components, specific types of damage or battery capacity retention. We also currently offer optional extended service plans that provide coverage beyond the new vehicle limited warranties for certain models in specified regions.

Energy Generation and Storage

We provide service and repairs to our energy product customers, including under warranty where applicable. We generally provide manufacturer's limited warranties with our energy storage products and offer certain extended limited warranties that are available at the time of purchase of the system. If we install a system, we also provide certain limited warranties on our installation workmanship.

For retrofit solar energy systems, we provide separate limited warranties for workmanship and against roof leaks, and for Solar Roof, we also provide limited warranties for defects and weatherization. For components not manufactured by us, we generally pass-through the applicable manufacturers' warranties.

As part of our solar energy system and energy storage contracts, we may provide the customer with performance guarantees that commit that the underlying system will meet or exceed the minimum energy generation or performance requirements specified in the contract.

Financial Services

Automotive

Purchase Financing and Leases

We offer leasing and/or loan financing arrangements for our vehicles in certain jurisdictions in North America, Europe and Asia ourselves and through various financial institutions. Under certain of such programs, we have provided resale value guarantees or buyback guarantees that may obligate us to cover a resale loss up to a certain limit or repurchase the subject vehicles at pre-determined values.

Insurance

In 2021, we launched our insurance product using real-time driving behavior in select states, which offers rates that are often better than other alternatives and promotes safer driving. Our insurance products are currently available in 12 states and we plan to expand the markets in which we offer insurance products, as part of our ongoing effort to decrease the total cost of ownership for our customers.

Energy Generation and Storage

We offer certain financing options to our solar customers, which enable the customer to purchase and own a solar energy system, Solar Roof or integrated solar and Powerwall system. Our solar PPAs, offered primarily to commercial customers, charge a fee per kilowatt-hour based on the amount of electricity produced by our solar energy systems.

Manufacturing

We currently have manufacturing facilities in the U.S. in Northern California, in Buffalo, New York, Gigafactory New York; in Austin, Texas, Gigafactory Texas and near Reno, Nevada, Gigafactory Nevada. At these facilities, we manufacture and assemble, among other things, vehicles, certain vehicle parts and components, such as our battery packs and battery cells, energy storage components and solar products and components.

Internationally, we also have manufacturing facilities in China (Gigafactory Shanghai) and Germany (Gigafactory Berlin-Brandenburg), which allows us to increase the affordability of our vehicles for customers in local markets by reducing transportation and manufacturing costs and eliminating the impact of unfavorable tariffs. In March 2023, we announced the location of our next Gigafactory in Monterrey, Mexico. Generally, we continue to expand production capacity at our existing facilities. We also intend to further increase cost-competitiveness in our significant markets by strategically adding local manufacturing.

Supply Chain

Our products use thousands of parts that are sourced from hundreds of suppliers across the world. We have developed close relationships with vendors of key parts such as battery cells, electronics and complex vehicle assemblies. Certain components purchased from these suppliers are shared or are similar across many product lines, allowing us to take advantage of pricing efficiencies from economies of scale.

As is the case for some automotive companies, some of our procured components and systems are sourced from single suppliers. Where multiple sources are available for certain key components, we work to qualify multiple suppliers for them where it is sensible to do so in order to minimize potential production risks due to disruptions in their supply. We also mitigate risk by maintaining safety stock for key parts and assemblies and die banks for components with lengthy procurement lead times.

Our products use various raw materials including aluminum, steel, cobalt, lithium, nickel and copper. Pricing for these materials is governed by market conditions and may fluctuate due to various factors outside of our control, such as supply and demand and market speculation. We strive to execute long-term supply contracts for such materials at competitive pricing when feasible, and we currently believe that we have adequate access to raw materials supplies to meet the needs of our operations.

Governmental Programs, Incentives and Regulations

Globally, the ownership of our products by our customers is impacted by various government credits, incentives, and policies. Our business and products are also subject to numerous governmental regulations that vary among jurisdictions.

The operation of our business is also impacted by various government programs, incentives, and other arrangements. See Note 2, *Summary of Significant Accounting Policies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Programs and Incentives

Inflation Reduction Act

On August 16, 2022, the Inflation Reduction Act of 2022 (“IRA”) was enacted into law and is effective for taxable years beginning after December 31, 2022, and remains subject to future guidance releases. The IRA includes multiple incentives to promote clean energy, electric vehicles, battery and energy storage manufacture or purchase, including through providing tax credits to consumers. For example, qualifying Tesla customers may receive up to $7,500 in federal tax credits for the purchase of qualified electric vehicles in the U.S. through 2032.

Automotive Regulatory Credits

We earn tradable credits in the operation of our business under various regulations related to zero-emission vehicles (“ZEVs”), greenhouse gas, fuel economy and clean fuel. We sell these credits to other regulated entities who can use the credits to comply with emission standards and other regulatory requirements. Sales of these credits are recognized within automotive regulatory credits revenue in our consolidated statements of operations included elsewhere in this Annual Report on Form 10-K.

Energy Storage System Incentives and Policies

While the regulatory regime for energy storage projects is still under development, there are various policies, incentives and financial mechanisms at the federal, state and local levels that support the adoption of energy storage.

For example, energy storage systems that are charged using solar energy may be eligible for the solar energy-related U.S. federal tax credits described below. The Federal Energy Regulatory Commission (“FERC”) has also taken steps to enable the participation of energy storage in wholesale energy markets. In addition, California and a number of other states have adopted procurement targets for energy storage, and behind-the-meter energy storage systems qualify for funding under the California Self Generation Incentive Program. Our customers primarily benefit directly under these programs. In certain instances our customers may transfer such credits to us as contract consideration. In such transactions, they are included as a component of energy generation and storage revenues in our consolidated statements of operations included elsewhere in this Annual Report on Form 10-K.

Pursuant to the IRA, under Sections 48, 48E and 25D of the Internal Revenue Code (”IRC”), standalone energy storage technology is eligible for a tax credit between 6% and 50% of qualified expenditures, regardless of the source of energy, which may be claimed by our customers for storage systems they purchase or by us for arrangements where we own the systems. These tax credits are primarily for the benefit of our customers and are currently scheduled to phase-out starting in 2032 or later.

Solar Energy System Incentives and Policies

U.S. federal, state and local governments have established various policies, incentives and financial mechanisms to reduce the cost of solar energy and to accelerate the adoption of solar energy. These incentives include tax credits, cash grants, tax abatements and rebates.

In particular, pursuant to the IRA, Sections 48, 48E and 25D of the IRC provides a tax credit between 6% and 70% of qualified commercial or residential expenditures for solar energy systems, which may be claimed by our customers for systems they purchase, or by us for arrangements where we own the systems for properties that meet statutory requirements. These tax credits are primarily for the direct benefit of our customers and are currently scheduled to phase-out starting in 2032 or later.

Regulations

Vehicle Safety and Testing

In the U.S., our vehicles are subject to regulation by the National Highway Traffic Safety Administration (“NHTSA”), including all applicable Federal Motor Vehicle Safety Standards (“FMVSS”) and the NHTSA bumper standard. Numerous FMVSS apply to our vehicles, such as crash-worthiness and occupant protection requirements. Our current vehicles fully comply and we expect that our vehicles in the future will fully comply with all applicable FMVSS with limited or no exemptions, however, FMVSS are subject to change from time to time. As a manufacturer, we must self-certify that our vehicles meet all applicable FMVSS and the NHTSA bumper standard, or otherwise are exempt, before the vehicles may be imported or sold in the U.S.

We are also required to comply with other federal laws administered by NHTSA, including the Corporate Average Fuel Economy standards, Theft Prevention Act requirements, labeling requirements and other information provided to customers in writing, Early Warning Reporting requirements regarding warranty claims, field reports, death and injury reports and foreign recalls, a Standing General Order requiring reports regarding crashes involving vehicles equipped with advanced driver assistance systems, and additional requirements for cooperating with compliance and safety investigations and recall reporting. The U.S. Automobile Information and Disclosure Act also requires manufacturers of motor vehicles to disclose certain information regarding the manufacturer’s suggested retail price, optional equipment and pricing. In addition, federal law requires inclusion of fuel economy ratings, as determined by the U.S. Department of Transportation and the Environmental Protection Agency (the “EPA”), and New Car Assessment Program ratings as determined by NHTSA, if available.

Our vehicles sold outside of the U.S. are subject to similar foreign compliance, safety, environmental and other regulations. Many of those regulations are different from those applicable in the U.S. and may require redesign and/or retesting. Some of those regulations impact or prevent the rollout of new vehicle features.

Self-Driving Vehicles

Generally, laws pertaining to self-driving vehicles are evolving globally, and in some cases may create restrictions on features or vehicle designs that we develop. While there are currently no federal U.S. regulations pertaining specifically to self-driving vehicles or self-driving equipment, NHTSA has published recommended guidelines on self-driving vehicles, apart from the FMVSS and manufacturer reporting obligations, and retains the authority to investigate and/or take action on the safety or compliance of any vehicle, equipment or features operating on public roads. Certain U.S. states also have legal restrictions on the operation, registration or licensure of self-driving vehicles, and many other states are considering them. This regulatory patchwork increases the legal complexity with respect to self-driving vehicles in the U.S.

In markets that follow the regulations of the United Nations Economic Commission for Europe (“ECE markets”), some requirements restrict the design of advanced driver-assistance or self-driving features, which can compromise or prevent their use entirely. Other applicable laws, both current and proposed, may hinder the path and timeline to introducing self-driving vehicles for sale and use in the markets where they apply.

Other key markets, including China, continue to consider self-driving regulation. Any implemented regulations may differ materially from the U.S. and ECE markets, which may further increase the legal complexity of self-driving vehicles and limit or prevent certain features.

Automobile Manufacturer and Dealer Regulation

In the U.S., state laws regulate the manufacture, distribution, sale and service of automobiles, and generally require motor vehicle manufacturers and dealers to be licensed in order to sell vehicles directly to residents. Certain states have asserted that the laws in such states do not permit automobile manufacturers to be licensed as dealers or to act in the capacity of a dealer, or that they otherwise restrict a manufacturer's ability to deliver or perform warranty repairs on vehicles. To sell vehicles to residents of states where we are not licensed as a dealer, we generally conduct the sale out of the state. In certain such states, we have opened "galleries" that serve an educational purpose and where sales may not occur.

Some automobile dealer trade associations have both challenged the legality of our operations in court and used administrative and legislative processes to attempt to prohibit or limit our ability to operate existing stores or expand to new locations. Certain dealer associations have also actively lobbied state licensing agencies and legislators to interpret existing laws or enact new laws in ways not favorable to our ownership and operation of our own retail and service locations. We expect such challenges to continue, and we intend to actively fight any such efforts.

Battery Safety and Testing

Our battery packs are subject to various U.S. and international regulations that govern transport of "dangerous goods," defined to include lithium-ion batteries, which may present a risk in transportation. We conduct testing to demonstrate our compliance with such regulations.

We use lithium-ion cells in our high voltage battery packs in our vehicles and energy storage products. The use, storage and disposal of our battery packs are regulated under existing laws and are the subject of ongoing regulatory changes that may add additional requirements in the future. We have agreements with third party battery recycling companies to recycle our battery packs, and we are also piloting our own recycling technology.

Solar Energy—General

We are subject to certain state and federal regulations applicable to solar and battery storage providers and sellers of electricity. To operate our systems, we enter into standard interconnection agreements with applicable utilities. Sales of electricity and non-sale equipment leases by third parties, such as our leases and PPAs, have faced regulatory challenges in some states and jurisdictions.

Solar Energy—Net Metering

Most states in the U.S. make net energy metering, or net metering, available to solar customers. Net metering typically allows solar customers to interconnect their solar energy systems to the utility grid and offset their utility electricity purchases by receiving a bill credit for excess energy generated by their solar energy system that is exported to the grid. In certain jurisdictions, regulators or utilities have reduced or eliminated the benefit available under net metering or have proposed to do so.

Competition

Automotive

The worldwide automotive market is highly competitive and we expect it will become even more competitive in the future as a significant and growing number of established and new automobile manufacturers, as well as other companies, have entered, or are reported to have plans to enter the electric vehicle market.

We believe that our vehicles compete in the market based on both their traditional segment classification as well as their propulsion technology. For example, Cybertruck competes with other pickup trucks, Model S and Model X compete primarily with premium sedans and premium SUVs and Model 3 and Model Y compete with small to medium-sized sedans and compact SUVs, which are extremely competitive markets. Competing products typically include internal combustion vehicles from more established automobile manufacturers; however, many established and new automobile manufacturers have entered or have announced plans to enter the market for electric and other alternative fuel vehicles. Overall, we believe these announcements and vehicle introductions, including the introduction of electric vehicles into rental car company fleets, promote the development of the electric vehicle market by highlighting the attractiveness of electric vehicles relative to the internal combustion vehicle. Many major automobile manufacturers have electric vehicles available today in major markets including the U.S., China and Europe, and other current and prospective automobile manufacturers are also developing electric vehicles. In addition, several manufacturers offer hybrid vehicles, including plug-in versions.

We believe that there is also increasing competition for our vehicle offerings as a platform for delivering self-driving technologies, charging solutions and other features and services, and we expect to compete in this developing market through continued progress on our Autopilot, FSD and neural network capabilities, Supercharger network and our infotainment offerings.

Energy Generation and Storage

Energy Storage Systems

The market for energy storage products is also highly competitive, and both established and emerging companies have introduced products that are similar to our product portfolio or that are alternatives to the elements of our systems. We compete with these companies based on price, energy density and efficiency. We believe that the specifications and features of our products, our strong brand and the modular, scalable nature of our energy storage products give us a competitive advantage in our markets.

Solar Energy Systems

The primary competitors to our solar energy business are the traditional local utility companies that supply energy to our potential customers. We compete with these traditional utility companies primarily based on price and the ease by which customers can switch to electricity generated by our solar energy systems. We also compete with solar energy companies that provide products and services similar to ours. Many solar energy companies only install solar energy systems, while others only provide financing for these installations. We believe we have a significant expansion opportunity with our offerings and that the regulatory environment is increasingly conducive to the adoption of renewable energy systems.

Intellectual Property

We place a strong emphasis on our innovative approach and proprietary designs which bring intrinsic value and uniqueness to our product portfolio. As part of our business, we seek to protect the underlying intellectual property rights of these innovations and designs such as with respect to patents, trademarks, copyrights, trade secrets, confidential information and other measures, including through employee and third-party nondisclosure agreements and other contractual arrangements. For example, we place a high priority on obtaining patents to provide the broadest and strongest possible protection to enable our freedom to operate our innovations and designs across all of our products and technologies as well as to protect and defend our product portfolio. We have also adopted a patent policy in which we irrevocably pledged that we will not initiate a lawsuit against any party for infringing our patents through activity relating to electric vehicles or related equipment for so long as such party is acting in good faith. We made this pledge in order to encourage the advancement of a common, rapidly-evolving platform for electric vehicles, thereby benefiting ourselves, other companies making electric vehicles and the world.

Environmental, Social and Governance (ESG) and Human Capital Resources

ESG

The very purpose of Tesla's existence is to accelerate the world's transition to sustainable energy. We believe the world cannot reduce carbon emissions without addressing both energy generation and consumption, and we are designing and manufacturing a complete energy and transportation ecosystem to achieve this goal. As we expand, we are building each new factory to be more efficient and sustainably designed than the previous one, including with respect to per-unit waste reduction and resource consumption, including water and energy usage. We are focused on further enhancing sustainability of operations outside of our direct control, including reducing the carbon footprint of our supply chain.

We are committed to sourcing only responsibly produced materials, and our suppliers are required to provide evidence of management systems that ensure social, environmental and sustainability best practices in their own operations, as well as to demonstrate a commitment to responsible sourcing into their supply chains. We have a zero-tolerance policy when it comes to child or forced labor and human trafficking by our suppliers and we look to the Organization for Economic Co-operation and Development Due Diligence Guidelines to inform our process and use feedback from our internal and external stakeholders to find ways to continually improve. We are also driving safety in our own factories by focusing on worker engagement. Our incidents per vehicle continue to drop even as our production volumes increase. We also strive to be an employer of choice by offering compelling, impactful jobs with best in-industry benefits.

We believe that sound corporate governance is critical to helping us achieve our goals, including with respect to ESG. We continue to evolve a governance framework that exercises appropriate oversight of responsibilities at all levels throughout the company and manages its affairs consistent with high principles of business ethics. Our ESG Sustainability Council is made up of leaders from across our company, and regularly presents to our Board of Directors, which oversees our ESG impacts, initiatives and priorities.

Human Capital Resources

A competitive edge for Tesla is its ability to attract and retain high quality employees. During the past year, Tesla made substantial investments in its workforce, further strengthening its standing as one of the most desirable and innovative companies to work for. As of December 31, 2023, our employee headcount worldwide was 140,473.

We have created an environment that fosters growth opportunities, and as of this report, nearly two-thirds (65%) of our managers were promoted from an internal, non-manager position, and 43% of our management employees have been with Tesla for more than five years. Tesla’s growth of 35% over the past two years has offered internal career development to our employees as well as the ability to make a meaningful contribution to a sustainable future.

We are able to retain our employees, in part, not only because employees can enjoy ownership in Tesla through stock (of which 89% have been given the opportunity to), but because we also provide them with excellent health benefits such as free counseling, paid parental leave, paid time off and zero-premium medical plan options that are made available on the first day of employment.

We recognize the positive impact that leaders can have on their teams and offer fundamental skills training and continuous development to all leaders through various programs globally.

We don’t stop there. Tesla has several other programs strategically designed to increase paths for greater career opportunity such as:

- **Technician Trainee (Service)** – The Tesla Technician Trainee Program provides on-the-job automotive maintenance training at Tesla, resulting in an industry certification. Targeted at individuals with limited experience, whether in industry or vocational schools, the program prepares trainees for employment as technicians. In 2023, we hired over 1,900 Technician Trainees across the U.S., Germany and China.

- **START (Manufacturing and Service)** – Tesla START is an intensive training program that complements the Technician Trainee program and equips individuals with the skills needed for a successful technician role at Tesla. We have partnered with colleges and technical academies to launch Tesla START in the U.S., United Kingdom and Germany. In 2023, we hired over 350 trainees for manufacturing and service roles through this program, providing an opportunity to transition into full-time employment.

- **Internships** – Annually, Tesla hires over 6,000 university and college students from around the world. We recruit from diverse student organizations and campuses, seeking top talent passionate about our mission. Our interns engage in meaningful work from day one, and we often offer them full-time positions post-internship.

- **Military Fellowship and Transition Programs –** The Military Fellowship and Transition Programs are designed to offer exiting military service members in the U.S. and Europe with career guidance on transitioning into the civil workforce. We partner with the career transition services of European Defence Ministries across five countries, as well as the U.S. Chamber of Commerce’s Hire our Heroes. These programs aim to convert high-performing individuals to full-time roles and create a veteran talent pipeline.

- **Apprenticeships** – Tesla Apprenticeships are offered globally, providing academic and on-the-job training to prepare specialists in skilled trades. Apprentices will complete between one to four years of on-the-job training. Apprentice programs have seen skilled trade hires across the U.S., Australia, Hong Kong, Korea and Germany.

- **Manufacturing Development Program** – Tesla's manufacturing pathway program is designed to provide graduating high school seniors with the financial resources, coursework and experience they need to start a successful manufacturing career at Tesla. We hired 373 graduates through this program in 2023, and our goal in 2024 is grow this program to over 600 students annually across our Fremont Factory, Gigafactory Nevada, Gigafactory Texas and Gigafactory New York.

- **Engineering Development Program** – Launched in January 2024, this program targets recent college and university graduates for specialized engineering fields. In collaboration with Austin Community College, the program educates early-career engineers in controls engineering, enhancing their knowledge of high-demand technologies for U.S. manufacturing.

We will continue to expand the opportunities for our employees to add skills and develop professionally with a new Employee Educational Assistance Program launching in the U.S. in the spring of 2024 to help employees pursue select certificates or degrees. With virtual, self-paced education options available, employees can pursue a new path or expand their knowledge while continuing to grow their career.

At Tesla, our employees show up passionate about making a difference in the world and for each other. We remain unwavering in our demand that our factories, offices, stores and service centers are places where our employees feel respected and appreciated. Our policies are designed to promote fairness and respect for everyone. We hire, evaluate and promote employees based on their skills and performance. Everyone is expected to be trustworthy, demonstrate excellence in their performance and collaborate with others. With this in mind, we will not tolerate certain behaviors. These include harassment, retaliation, violence, intimidation and discrimination of any kind on the basis of race, color, religion, national origin, gender, sexual orientation, gender identity, gender expression, age, disability or veteran status.

Anti-harassment training is conducted on day one of new hire orientation for all employees and reoccurring for leaders. In addition, we run various leadership development programs throughout the year aimed at enhancing leaders' skills, and in particular, helping them to understand how to appropriately respond to and address employee concerns.

Employees are encouraged to speak up both in regard to misconduct and safety concerns and can do so by contacting the integrity line, submitting concerns through our Take Charge process, or notifying their Human Resource Partner or any member of management. Concerns are reviewed in accordance with established protocols by investigators with expertise, who also review for trends and outcomes for remediation and appropriate controls. Responding to questions timely is key so Human Resource Partners for each functional area are visible throughout facilities and are actively involved in driving culture and engagement alongside business leaders.

Available Information

We file or furnish periodic reports and amendments thereto, including our Annual Reports on Form 10-K, our Quarterly Reports on Form 10-Q and Current Reports on Form 8-K, proxy statements and other information with the SEC. In addition, the SEC maintains a website (www.sec.gov) that contains reports, proxy and information statements, and other information regarding issuers that file electronically. Our website is located at www.tesla.com, and our reports, amendments thereto, proxy statements and other information are also made available, free of charge, on our investor relations website at ir.tesla.com as soon as reasonably practicable after we electronically file or furnish such information with the SEC. The information posted on our website is not incorporated by reference into this Annual Report on Form 10-K.

ITEM 1A. RISK FACTORS

You should carefully consider the risks described below together with the other information set forth in this report, which could materially affect our business, financial condition and future results. The risks described below are not the only risks facing our company. Risks and uncertainties not currently known to us or that we currently deem to be immaterial also may materially adversely affect our business, financial condition and operating results.

Risks Related to Our Ability to Grow Our Business

We may experience delays in launching and ramping the production of our products and features, or we may be unable to control our manufacturing costs.

We have previously experienced and may in the future experience launch and production ramp delays for new products and features. For example, we encountered unanticipated supplier issues that led to delays during the initial ramp of our first Model X and experienced challenges with a supplier and with ramping full automation for certain of our initial Model 3 manufacturing processes. In addition, we may introduce in the future new or unique manufacturing processes and design features for our products. As we expand our vehicle offerings and global footprint, there is no guarantee that we will be able to successfully and timely introduce and scale such processes or features.

In particular, our future business depends in large part on increasing the production of mass-market vehicles. In order to be successful, we will need to implement, maintain and ramp efficient and cost-effective manufacturing capabilities, processes and supply chains and achieve the design tolerances, high quality and output rates we have planned at our manufacturing facilities in California, Nevada, Texas, China, Germany and any future sites such as Mexico. We will also need to hire, train and compensate skilled employees to operate these facilities. Bottlenecks and other unexpected challenges such as those we experienced in the past may arise during our production ramps, and we must address them promptly while continuing to improve manufacturing processes and reducing costs. If we are not successful in achieving these goals, we could face delays in establishing and/or sustaining our product ramps or be unable to meet our related cost and profitability targets.

We have experienced, and may also experience similar future delays in launching and/or ramping production of our energy storage products and Solar Roof; new product versions or variants; new vehicles; and future features and services based on artificial intelligence. Likewise, we may encounter delays with the design, construction and regulatory or other approvals necessary to build and bring online future manufacturing facilities and products.

Any delay or other complication in ramping the production of our current products or the development, manufacture, launch and production ramp of our future products, features and services, or in doing so cost-effectively and with high quality, may harm our brand, business, prospects, financial condition and operating results.

Our suppliers may fail to deliver components according to schedules, prices, quality and volumes that are acceptable to us, or we may be unable to manage these components effectively.

Our products contain thousands of parts purchased globally from hundreds of suppliers, including single-source direct suppliers, which exposes us to multiple potential sources of component shortages. Unexpected changes in business conditions, materials pricing, including inflation of raw material costs, labor issues, wars, trade policies, natural disasters, health epidemics such as the global COVID-19 pandemic, trade and shipping disruptions, port congestions, cyberattacks and other factors beyond our or our suppliers' control could also affect these suppliers' ability to deliver components to us or to remain solvent and operational. For example, a global shortage of semiconductors beginning in early 2021 has caused challenges in the manufacturing industry and impacted our supply chain and production. Additionally, if our suppliers do not accurately forecast and effectively allocate production or if they are not willing to allocate sufficient production to us, or face other challenges such as insolvency, it may reduce our access to components and require us to search for new suppliers. The unavailability of any component or supplier could result in production delays, idle manufacturing facilities, product design changes and loss of access to important technology and tools for producing and supporting our products, as well as impact our capacity expansion and our ability to fulfill our obligations under customer contracts. Moreover, significant increases in our production or product design changes by us have required and may in the future require us to procure additional components in a short amount of time. We have faced in the past, and may face suppliers who are unwilling or unable to sustainably meet our timelines or our cost, quality and volume needs, which may increase our costs or require us to replace them with other sources. Finally, as we construct new manufacturing facilities and add production lines to existing facilities, we may experience issues in correspondingly increasing the level of localized procurement at those facilities. While we believe that we will be able to secure additional or alternate sources or develop our own replacements for most of our components, there is no assurance that we will be able to do so quickly or at all. Additionally, we may be unsuccessful in our continuous efforts to negotiate with existing suppliers to obtain cost reductions and avoid unfavorable changes to terms, source less expensive suppliers for certain parts and redesign certain parts to make them less expensive to produce, especially in the case of increases in materials pricing. Any of these occurrences may harm our business, prospects, financial condition and operating results.

As the scale of our vehicle production increases, we will also need to accurately forecast, purchase, warehouse and transport components at high volumes to our manufacturing facilities and servicing locations internationally. If we are unable to accurately match the timing and quantities of component purchases to our actual needs or successfully implement automation, inventory management and other systems to accommodate the increased complexity in our supply chain and parts management, we may incur unexpected production disruption, storage, transportation and write-off costs, which may harm our business and operating results.

We may be unable to meet our projected construction timelines, costs and production ramps at new factories, or we may experience difficulties in generating and maintaining demand for products manufactured there.

Our ability to increase production of our vehicles on a sustained basis, make them affordable globally by accessing local supply chains and workforces and streamline delivery logistics is dependent on the construction and ramp of our current and future factories. The construction of and commencement and ramp of production at these factories are subject to a number of uncertainties inherent in all new manufacturing operations, including ongoing compliance with regulatory requirements, procurement and maintenance of construction, environmental and operational licenses and approvals for additional expansion, supply chain constraints, hiring, training and retention of qualified employees and the pace of bringing production equipment and processes online with the capability to manufacture high-quality units at scale. Moreover, we will have to establish and ramp production of our proprietary battery cells and packs at our new factories, and we additionally intend to incorporate sequential design and manufacturing changes into vehicles manufactured at each new factory. If we experience any issues or delays in meeting our projected timelines, costs, capital efficiency and production capacity for our new factories, expanding and managing teams to implement iterative design and production changes there, maintaining and complying with the terms of any debt financing that we obtain to fund them or generating and maintaining demand for the vehicles we manufacture there, our business, prospects, operating results and financial condition may be harmed.

We may be unable to grow our global product sales, delivery and installation capabilities and our servicing and vehicle charging networks, or we may be unable to accurately project and effectively manage our growth.

Our success will depend on our ability to continue to expand our sales capabilities. We are targeting a global mass demographic with a broad range of potential customers, in which we have relatively limited experience projecting demand and pricing our products. We currently produce numerous international variants at a limited number of factories, and if our specific demand expectations for these variants prove inaccurate, we may not be able to timely generate deliveries matched to the vehicles that we produce in the same timeframe or that are commensurate with the size of our operations in a given region. Likewise, as we develop and grow our energy products and services worldwide, our success will depend on our ability to correctly forecast demand in various markets.

Because we do not have independent dealer networks, we are responsible for delivering all of our vehicles to our customers. As our production volumes continue to grow, we have faced in the past, and may face challenges with deliveries at increasing volumes, particularly in international markets requiring significant transit times. We have also deployed a number of delivery models, such as deliveries to customers' homes and workplaces and touchless deliveries, but there is no guarantee that such models will be scalable or be accepted globally. Likewise, as we ramp our energy products, we are working to substantially increase our production and installation capabilities. If we experience production delays or inaccurately forecast demand, our business, financial condition and operating results may be harmed.

Moreover, because of our unique expertise with our vehicles, we recommend that our vehicles be serviced by us or by certain authorized professionals. If we experience delays in adding servicing capacity or servicing our vehicles efficiently, or experience unforeseen issues with the reliability of our vehicles, particularly higher-volume additions to our fleet such as Model 3 and Model Y, it could overburden our servicing capabilities and parts inventory. Similarly, the increasing number of Tesla vehicles also requires us to continue to rapidly increase the number of our Supercharger stations and connectors throughout the world.

There is no assurance that we will be able to ramp our business to meet our sales, delivery, installation, servicing and vehicle charging targets globally, that our projections on which such targets are based will prove accurate or that the pace of growth or coverage of our customer infrastructure network will meet customer expectations. These plans require significant cash investments and management resources and there is no guarantee that they will generate additional sales or installations of our products, or that we will be able to avoid cost overruns or be able to hire additional personnel to support them. As we expand, we will also need to ensure our compliance with regulatory requirements in various jurisdictions applicable to the sale, installation and servicing of our products, the sale or dispatch of electricity related to our energy products and the operation of Superchargers. If we fail to manage our growth effectively, it may harm our brand, business, prospects, financial condition and operating results.

We will need to maintain and significantly grow our access to battery cells, including through the development and manufacture of our own cells, and control our related costs.

We are dependent on the continued supply of lithium-ion battery cells for our vehicles and energy storage products, and we will require substantially more cells to grow our business according to our plans. Currently, we rely on suppliers such as Panasonic and Contemporary Amperex Technology Co. Limited (CATL) for these cells. We have to date fully qualified only a very limited number of such suppliers and have limited flexibility in changing suppliers. Any disruption in the supply of battery cells from our suppliers could limit production of our vehicles and energy storage products. In the long term, we intend to supplement cells from our suppliers with cells manufactured by us, which we believe will be more efficient, manufacturable at greater volumes and more cost-effective than currently available cells. However, our efforts to develop and manufacture such battery cells have required, and may continue to require, significant investments, and there can be no assurance that we will be able to achieve these targets in the timeframes that we have planned or at all. If we are unable to do so, we may have to curtail our planned vehicle and energy storage product production or procure additional cells from suppliers at potentially greater costs, either of which may harm our business and operating results.

In addition, the cost and mass production of battery cells, whether manufactured by our suppliers or by us, depends in part upon the prices and availability of raw materials such as lithium, nickel, cobalt and/or other metals. The prices for these materials fluctuate and their available supply may be unstable, depending on market conditions and global demand for these materials. For example, as a result of increased global production of electric vehicles and energy storage products, suppliers of these raw materials may be unable to meet our volume needs. Additionally, our suppliers may not be willing or able to reliably meet our timelines or our cost and quality needs, which may require us to replace them with other sources. Any reduced availability of these materials may impact our access to cells and our growth, and any increases in their prices may reduce our profitability if we cannot recoup such costs through increased prices. Moreover, our inability to meet demand and any product price increases may harm our brand, growth, prospects and operating results.

Our future growth and success are dependent upon consumers' demand for electric vehicles and specifically our vehicles in an automotive industry that is generally competitive, cyclical and volatile.

Though we continue to see increased interest and adoption of electric vehicles, if the market for electric vehicles in general and Tesla vehicles in particular does not develop as we expect, develops more slowly than we expect, or if demand for our vehicles decreases in our markets or our vehicles compete with each other, our business, prospects, financial condition and operating results may be harmed.

In addition, electric vehicles still constitute a small percentage of overall vehicle sales. As a result, the market for our vehicles could be negatively affected by numerous factors, such as:

- perceptions about electric vehicle features, quality, safety, performance and cost;
- perceptions about the limited range over which electric vehicles may be driven on a single battery charge, and access to charging facilities;
- competition, including from other types of alternative fuel vehicles, plug-in hybrid electric vehicles and high fuel-economy internal combustion engine vehicles;
- volatility in the cost of oil, gasoline and energy;
- government regulations and economic incentives and conditions; and
- concerns about our future viability.

The target demographics for our vehicles are highly competitive. Sales of vehicles in the automotive industry tend to be cyclical in many markets, which may expose us to further volatility. We also cannot predict the duration or direction of current global trends or their sustained impact on consumer demand. Ultimately, we continue to monitor macroeconomic conditions to remain flexible and to optimize and evolve our business as appropriate, and attempt to accurately project demand and infrastructure requirements globally and deploy our production, workforce and other resources accordingly. Rising interest rates may lead to consumers to increasingly pull back spending, including on our products, which may harm our demand, business and operating results. If we experience unfavorable global market conditions, or if we cannot or do not maintain operations at a scope that is commensurate with such conditions or are later required to or choose to suspend such operations again, our business, prospects, financial condition and operating results may be harmed.

We face strong competition for our products and services from a growing list of established and new competitors.

The worldwide automotive market is highly competitive today and we expect it will become even more so in the future. A significant and growing number of established and new automobile manufacturers, as well as other companies, have entered, or are reported to have plans to enter, the market for electric and other alternative fuel vehicles, including hybrid, plug-in hybrid and fully electric vehicles, as well as the market for self-driving technology and other vehicle applications and software platforms. In some cases, our competitors offer or will offer electric vehicles in important markets such as China and Europe, and/or have announced an intention to produce electric vehicles exclusively at some point in the future. In addition, certain government and economic incentives which provide benefits to manufacturers who assemble domestically or have local suppliers, may provide a greater benefit to our competitors, which could negatively impact our profitability. Many of our competitors have significantly more or better-established resources than we do to devote to the design, development, manufacturing, distribution, promotion, sale and support of their products. Increased competition could result in our lower vehicle unit sales, price reductions, revenue shortfalls, loss of customers and loss of market share, which may harm our business, financial condition and operating results.

We also face competition in our energy generation and storage business from other manufacturers, developers, installers and service providers of competing energy technologies, as well as from large utilities. Decreases in the retail or wholesale prices of electricity from utilities or other renewable energy sources could make our products less attractive to customers and lead to an increased rate of customer defaults.

Risks Related to Our Operations

We may experience issues with lithium-ion cells or other components manufactured at our Gigafactories, which may harm the production and profitability of our vehicle and energy storage products.

Our plan to grow the volume and profitability of our vehicles and energy storage products depends on significant lithium-ion battery cell production, including by our partner Panasonic at Gigafactory Nevada. We also produce several vehicle components at our Gigafactories, such as battery modules and packs and drive units, and manufacture energy storage products. If we are unable to or otherwise do not maintain and grow our respective operations, or if we are unable to do so cost-effectively or hire and retain highly-skilled personnel there, our ability to manufacture our products profitably would be limited, which may harm our business and operating results.

Finally, the high volumes of lithium-ion cells and battery modules and packs manufactured by us and by our suppliers are stored and recycled at our various facilities. Any mishandling of these products may cause disruption to the operation of such facilities. While we have implemented safety procedures related to the handling of the cells, there can be no assurance that a safety issue or fire related to the cells would not disrupt our operations. Any such disruptions or issues may harm our brand and business.

We face risks associated with maintaining and expanding our international operations, including unfavorable and uncertain regulatory, political, economic, tax and labor conditions.

We are subject to legal and regulatory requirements, political uncertainty and social, environmental and economic conditions in numerous jurisdictions, including markets in which we generate significant sales, over which we have little control and which are inherently unpredictable. Our operations in such jurisdictions, particularly as a company based in the U.S., create risks relating to conforming our products to regulatory and safety requirements and charging and other electric infrastructures; organizing local operating entities; establishing, staffing and managing foreign business locations; attracting local customers; navigating foreign government taxes, regulations and permit requirements; enforceability of our contractual rights; trade restrictions, customs regulations, tariffs and price or exchange controls; and preferences in foreign nations for domestically manufactured products. For example, we monitor tax legislation changes on a global basis, including changes arising as a result of the Organization for Economic Cooperation and Development’s multi-jurisdictional plan of action to address base erosion and profit shifting. Such conditions may increase our costs, impact our ability to sell our products and require significant management attention, and may harm our business if we are unable to manage them effectively.

Our business may suffer if our products or features contain defects, fail to perform as expected or take longer than expected to become fully functional.

If our products contain design or manufacturing defects, whether relating to our software or hardware, that cause them not to perform as designed or intended or that require repair, or certain features of our vehicles such as new Autopilot or FSD Capability features take longer than expected to become enabled, are legally restricted or become subject to onerous regulation, our ability to develop, market and sell our products and services may be harmed, and we may experience delivery delays, product recalls, allegations of product liability, breach of warranty and related consumer protection claims and significant warranty and other expenses. While we are continuously working to develop and improve our products’ capability and performance, there is no guarantee that any incremental changes in the specific software or equipment we deploy in our vehicles over time will not result in initial functional disparities from prior iterations or will perform as forecast in the timeframe we anticipate, or at all. Although we attempt to remedy any issues we observe in our products as effectively and rapidly as possible, such efforts may not be timely, may hamper production or may not completely satisfy our customers. We have performed, and continue to perform, extensive internal testing on our products and features, though, like the rest of the industry, we currently have a limited frame of reference by which to evaluate certain aspects of their long-term quality, reliability, durability and performance characteristics, including exposure to or consequence of external attacks. While we attempt to identify and address or remedy defects we identify pre-production and sale, there may be latent defects that we may be unable to detect or control for in our products, and thereby address, prior to their sale to or installation for customers.

We may be required to defend or insure against product liability claims.

The automobile industry generally experiences significant product liability claims, and as such we face the risk of such claims in the event our vehicles do not perform or are claimed to not have performed as expected. As is true for other automakers, our vehicles have been involved and we expect in the future will be involved in accidents resulting in death or personal injury, and such accidents where Autopilot, Enhanced Autopilot or FSD Capability features are engaged are the subject of significant public attention, especially in light of NHTSA's Standing General Order requiring reports regarding crashes involving vehicles with advanced driver assistance systems. We have experienced, and we expect to continue to face, claims and regulatory scrutiny arising from or related to misuse or claimed failures or alleged misrepresentations of such new technologies that we are pioneering. In addition, the battery packs that we produce make use of lithium-ion cells. On rare occasions, lithium-ion cells can rapidly release the energy they contain by venting smoke and flames in a manner that can ignite nearby materials as well as other lithium-ion cells. While we have designed our battery packs to passively contain any single cell's release of energy without spreading to neighboring cells, there can be no assurance that a field or testing failure of our vehicles or other battery packs that we produce will not occur, in particular due to a high-speed crash. Likewise, as our solar energy systems and energy storage products generate and store electricity, they have the potential to fail or cause injury to people or property. Any product liability claim may subject us to lawsuits and substantial monetary damages, product recalls or redesign efforts, and even a meritless claim may require us to defend it, all of which may generate negative publicity and be expensive and time-consuming. In most jurisdictions, we generally self-insure against the risk of product liability claims for vehicle exposure, meaning that any product liability claims will likely have to be paid from company funds and not by insurance.

We will need to maintain public credibility and confidence in our long-term business prospects in order to succeed.

In order to maintain and grow our business, we must maintain credibility and confidence among customers, suppliers, analysts, investors, ratings agencies and other parties in our long-term financial viability and business prospects. Maintaining such confidence may be challenging due to our limited operating history relative to established competitors; customer unfamiliarity with our products; any delays we may experience in scaling manufacturing, delivery and service operations to meet demand; competition and uncertainty regarding the future of electric vehicles or our other products and services; our quarterly production and sales performance compared with market expectations; and other factors including those over which we have no control. In particular, Tesla's products, business, results of operations, and statements and actions of Tesla and its management are subject to significant amounts of commentary by a range of third parties. Such attention can include criticism, which may be exaggerated or unfounded, such as speculation regarding the sufficiency or stability of our management team. Any such negative perceptions, whether caused by us or not, may harm our business and make it more difficult to raise additional funds if needed.

We may be unable to effectively grow, or manage the compliance, residual value, financing and credit risks related to, our various financing programs.

We offer financing arrangements for our vehicles in North America, Europe and Asia primarily ourselves and through various financial institutions. We also currently offer vehicle financing arrangements directly through our local subsidiaries in certain markets. Depending on the country, such arrangements are available for specified models and may include operating leases directly with us under which we typically receive only a very small portion of the total vehicle purchase price at the time of lease, followed by a stream of payments over the term of the lease. We have also offered various arrangements for customers of our solar energy systems whereby they pay us a fixed payment to lease or finance the purchase of such systems or purchase electricity generated by them. If we do not successfully monitor and comply with applicable national, state and/or local financial regulations and consumer protection laws governing these transactions, we may become subject to enforcement actions or penalties.

The profitability of any directly-leased vehicles returned to us at the end of their leases depends on our ability to accurately project our vehicles' residual values at the outset of the leases, and such values may fluctuate prior to the end of their terms depending on various factors such as supply and demand of our used vehicles, economic cycles and the pricing of new vehicles. We have made in the past and may make in the future certain adjustments to our prices from time to time in the ordinary course of business, which may impact the residual values of our vehicles and reduce the profitability of our vehicle leasing program. The funding and growth of this program also rely on our ability to secure adequate financing and/or business partners. If we are unable to adequately fund our leasing program through internal funds, partners or other financing sources, and compelling alternative financing programs are not available for our customers who may expect or need such options, we may be unable to grow our vehicle deliveries. Furthermore, if our vehicle leasing business grows substantially, our business may suffer if we cannot effectively manage the resulting greater levels of residual risk.